Exercises, sheet 10. Let $K \models \mathrm{DCF}_0$ and $k \subseteq K$ be a differential subfield.

1. Solve Exercise 15.2.
2. Specify precisely the uniqueness from Theorem 15.8.
3. Show that k-Kolchin closed sets form closed sets of some topology on K^n.
4. Prove Corollary 15.16.
5. Prove Lemma 15.17.
6. Prove the algebro-geometric Nullstellensatz using quantifier elimination in ACF_0.
7. Show that every definable subset of the constants C is either finite or co-finite in C (constants are strongly minimal).
8. Consider the differential equation $X' = aX$ with a parameter $a \in k$. Let $s \in K$ be a solution (why does it exist?). Show that all solutions to this equation are of the form $c \cdot s$ for some constant $c \in C$.